

Carlos Armando Aguila

caguila1998@gmail.com | aguilac@seas.upenn.edu | [LinkedIn](#)

INTERESTS

Passionate about pioneering at the intersection of BioTech innovation and healthcare, I am driven to translate cutting-edge scientific discoveries into solutions that meaningfully impact patient care. Technically, my research develops novel computational approaches, utilizing advanced signal processing and machine learning, to analyze intracranial EEG data in epilepsy patients, improving surgical planning and providing new insights into the epileptic network. I aim to leverage my deep technical expertise alongside my entrepreneurial mindset to bridge the critical gap between groundbreaking research and accessible, market-ready healthcare technologies.

EDUCATION

PhD	Bioengineering, University of Pennsylvania Dissertation: "Mapping dynamic epileptic networks using spontaneous and evoked interictal EEG" Advisors: Brian Litt, MD & Erin C. Conrad, MD	2021 - Present
MS	Bioengineering, University of Pennsylvania	2021 - 2025
BS	Biomedical Engineering, University of Miami GPA: 3.91/4.0, Magna Cum Laude	2016 - 2020

PROFESSIONAL AND RESEARCH EXPERIENCE

Center for Neuroengineering and Therapeutics, University of Pennsylvania <i>Graduate Research Fellow</i> , Advisor: Erin C. Conrad, MD & Brian Litt, MD	Philadelphia, PA 2022 - Present
- Conducted time-series analysis (calculate spike rates, multidien spike cycles, networks, etc.) of intracranial electroencephalography (EEG) data across a multi-center cohort of patients. - Developed machine learning models to integrate interictal and ictal features to predict mesial temporal lobe epilepsy. - Designed an automated interictal spike detection algorithm with a robust morphology annotation system for comprehensive interictal analysis that guides clinician decision-making and improves surgical outcomes. - Constructed an automated analysis pipeline for rigorously characterizing epileptogenic brain networks from cortical stimulation trials utilized in epilepsy care by quantifying stimulation-evoked responses. - Collaborated with graduate students to characterize ictal intracranial EEG using a novel unsupervised deep learning network to cluster similar seizure dynamics across patients. - Developed an automated analysis pipeline to study the effects of network perturbation on evoked epileptiform discharges to better guide clinical neurostimulation and understand the underlying epileptic network.	
Penn Biotech Group, University of Pennsylvania <i>Consultant</i> , Supervisor: Ana Defendini, PhD	Philadelphia, PA 2022 - Present
- Carried out comprehensive analyses of academic technologies to assess commercial viability and facilitate commercialization of promising ones. - Conducted in-depth market landscape research, outlining where a technology might fit within the commercialization pipeline and the industry environment - Drafted invention assessments, detailing a foundational understanding of a technology and commercializing potential. - Delivering sound advice to company founders to ensure their success.	
The Neural Stack <i>Founder & Author</i>	Philadelphia, PA 2025 - Present
- Authoring a series of technical deep dives analyzing the commercial viability, technological hurdles, and key innovators within the Brain-Computer Interfaces sector. - Translating complex neurotechnology concepts into strategic insights for dissemination to venture capital investors, founders, and industry stakeholders.	

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Physiomimetic Microsystems Laboratory (PML), University of Miami

Miami, FL

Research Assistant, Advisor: Ashutosh Agarwal, PhD

2019 - 2021

- Collaborated with graduate students designing and creating platforms to better study shear stress, oxygen diffusion in pancreatic islet organ-on-chip platforms, and metastatic cancer on an extravasation model.
- Iteratively developed optimized-protocols to enhance quality of organ-on-chip platforms during micro-fabrication.

Vice President's Division of Student Affairs, University of Miami

Miami, FL

Student Analyst, Advisor: Patricia Whiteley, PhD

2018 - 2020

- Created early iteration of real-time, self-updating student database to consolidate information and ease access.
- Wrote financial reports to visualize large volumes of financial information clearly and intuitively.
- Served as the first point of contact for students, parents, and administrators, efficiently triaging inquiries and resolving issues independently to minimize escalation to the Vice President.

Department of Psychiatry and Behavioral Sciences, University of Miami

Miami, FL

Research Assistant, Supervisor: Deborah Weiss, PhD

Summer 2017

- Analyzed and automated translation of audio interviews from Zambian clinics to evaluate HIV prevention clinic effectiveness on prevention of HIV and other sexually transmitted diseases in Africa.
- Conducted extensive research on the impact of methamphetamine abuse and HIV on thyroid function among men in South Florida, exploring relationships amongst demographic, psychosocial, and physiological factors.

Siemens South Miami

Miami, FL

Intern, Supervisor: Guillermo Padron

Summer 2017

- Gained hands-on experience with various Siemens medical imaging systems including MRI, CT scanners, and diagnostic equipment.
- Assisted with preventative maintenance procedures and witnessed corrective maintenance techniques on critical healthcare systems.

AWARDS/HONORS

NSF Graduate Research Fellowship Program	2023
NIH R01 Diversity Supplement	2023
University of Pennsylvania Fontaine Fellow	2021
University of Miami - Alpha Eta Mu Beta Award	2020
Alpha Eta Mu Beta – National biomedical engineering honor society	2019
Omicron Delta Kappa Honor Society – National leadership honor society	2019
Tau Beta Pi Honor Society – National engineering honor society	2019
Florida-Georgia Louis Stokes Alliance for Minority Participation	2016
University of Miami Ronald A. Hammond Scholarship	2016

SKILLS

Machine Learning and Data Science: Clustering, Unsupervised/Semi-Supervised/Supervised Learning, Deep Learning (Artificial Neural Networks), Time Series Analysis, Image Analysis

Data Science Software: Scikit-Learn, Scipy, Statsmodels, Pandas, Tensorflow

Programming: Highly Proficient in: Python, MATLAB

Comfortable with: HTML, CSS, R

Familiar with: C, C++, Java

High Performance Computing: Unix, Linux, Remote Super Computing Clusters, Code Parallelization (Python, MATLAB), GPU Computing, git Version Control

Engineering Design: CAD, SOLIDWORKS, Rapid Prototyping

Software: Arduino IDE, COMSOL Multiphysics, ImageJ

Document Preparation and Graphic Design: Microsoft Office, Adobe Illustrator, Affinity Designer

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Languages: Native in Spanish, fluent in English

LEADERSHIP & MENTORING

Camp Kesem

Currently: Graduate Advisory Board

2017 - Present

- Enabled fundraising of over \$120,000 for a camp for kids to cope with parental cancer.
- Assisted in preparing the annual Make the Magic Gala, the largest fundraiser the organization holds – raising over \$20,000.
- Identified and researched potential corporate and institutional sponsors, expanding fundraising – allowing more kids to participate in the annual retreat.
- Supervised twenty high school boys whose parents had been affected by cancer as a counselor. Organized activities including rock climbing and facilitated nightly cabin chats and empowerment ceremonies to create a supportive environment for campers to process their experiences.

Nucleate, Philadelphia Chapter

Philadelphia, PA

Co-Head Technology and Partnership

2025 - Present

- Engage directly with Philadelphia-area technology transfer offices and inventors to recruit early-stage technologies for Nucleate Philadelphia's 2025 Activator cohort.
- Cultivated and managed a regional partnership portfolio of VCs, law firms, and life science companies to secure financial/in-kind support, mentorship, and resources for Philadelphia's academic-led biotech ventures.

Penn Undergraduate Research Mentoring Program, University of Pennsylvania

Philadelphia, PA

Mentor

2023 – Present

- Lead first-year undergraduate students through a new research opportunity, helping them formulate robust scientific hypotheses and designing effective experimental protocols.
- Provide hands-on training in data analysis techniques in Python and MATLAB to enhance student's technical skills.
- Guide students through literature reviews, data collection, and time-series analysis. Helping them develop critical thinking and scientific writing skills.

iPraxis, University of Pennsylvania

Philadelphia, PA

Mentor

2021 – 2023

- Judged and mentored science fair projects across 4 Philadelphia K-8 schools, teaching new scientific skills and the scientific process.
- Conducted hands-on science demonstrations teaching pH and polymer concepts to a classroom of students from Cook-Wissahickon elementary school.

Road2Research, University of Pennsylvania

Philadelphia, PA

Mentor

2021 – 2023

- Provide comprehensive mentorship to three underserved high school students, guiding them through hands-on research experiences and academic development.
- Share personal academic journeys to inspire first-generation and underrepresented students to pursue careers in STEM fields

Inspire U, University of Miami

Miami, FL

Leadership Team & Mentor

2016 – 2020

- Served on leadership team organizing monthly pre-college workshops for first-generation underrepresented high school students from Booker T. Washington Senior High School
- Mentored disadvantaged students on college preparation, resulting in mentees gaining acceptance to elite institutions including Notre Dame.

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- Established peer mentoring connections between college and high school students, contributing to over 50 high school graduates pursuing higher education.

Patient Navigator, University of Miami

Miami, FL

Volunteer

2018 - 2020

- Identified high risk patients at the Miller School of Medicine's Health Fair, and assisted them on attaining health insurance, necessary care, and meeting their health goals.

Engineering Student Council, University of Miami

Miami, FL

Event Chair

2019 – 2020

- Organized E-week 2020 for the College of Engineering at the University of Miami, a week full of orchestrating engineering exhibitions to facilitate student involvement with different engineering clubs on campus.

Cuban Order of the Knights of Malta & ILAC Medical Mission

Miami, FL

Medical Volunteer

February 2019

- Collaborated with international medical professionals to facilitate care for patients requiring advanced treatment in a small village called Licey al Medio, Dominican Republic.
- Observed firsthand the critical need for affordable, accessible medical devices in underserved regions.
- Provided scribing support for over 100 patients daily during 11+ hour shifts.

"Snips for Autism" Documentary, University of Miami

Miami, FL

Documentary Creator

Spring 2019

- Produced and directed documentary highlighting sensory sensitivity challenges faced by children with autism.
- Followed and documented a family's experience preparing their child with autism for haircut experiences.
- Successfully published documentary through South Florida PBS.

JOURNAL PUBLICATIONS

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- CA Aguila, Z Zhou, ..., EC Conrad (2025). Low frequency stimulation has heterogeneous effects on interictal spike rates. (*in preparation*).
- CA Aguila, A Lucas, SB Lavelle, ..., B Litt, EC Conrad. (2025). Mesial-to-lateral patterns of epileptiform activity identify the seizure onset zone in mesial temporal lobe epilepsy. *Epilepsia*.
- Z Goldblum, H Shi, CA Aguila, ..., B Litt. (2025). An Implantable Device that Converses with Patients and Learns to Co-Manage Epilepsy. (*in submission*).
- S Lavelle, WKS Ojemann, CA Aguila, ..., N Sinha. (2025). Models of Cortico-cortical Evoked Potentials Reveal a Distance, State and Epileptogenicity Dependent Relationship. (*in preparation*).
- E Conrad, CA Aguila, ..., CA Ellis. (2025) Automated estimation of seizure and spike burden and their association in a large epilepsy cohort. (*in preparation*).
- J Kim, CA Aguila, ..., EC Conrad (2025). Interictal Spike Asymmetry on Scalp EEG to Lateralize Temporal Lobe Epilepsy. (*in submission*).
- AR Pattnaik, Z Xu, CA Aguila, ..., B Litt. (2025). The seizure embedding map: a spatio-temporal transformer for characterizing ictal onset in intracranial EEG at scale. *Brain*. (Under review)
- AR Pattnaik, IZ Ong, CA Aguila, ..., B Litt (2025). Normative Intracranial EEG Localizes Epileptic Abnormalities Across States of Wakefulness. *Annals of Neurology*.
- N Ghosn, K Xie, CA Aguila, ..., B Litt (2025). Inter-ictal spikes are not modulated by anti-seizure medication taper in the epilepsy monitoring unit: a tale of two confounders. *Epilepsia*. (*in submission*)
- N Ghosn, D Ma, CA Aguila, ..., B Litt (2025). Synchrony is a robust iEEG biomarker for antiseizure medication load in epileptic patients. (*in preparation*).
- WKS Ojemann, Z Xu, CA Aguila, ..., EC Conrad (2025). Neural dynamic divergence: an unsupervised machine learning approach to modelling ictal state transitions. (*in submission*).

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- EC Conrad, Alfredo Lucas, **CA Aguila**, ..., B Litt (2024). Interictal intracranial EEG asymmetry lateralizes temporal lobe epilepsy. *Brain Communications*.
- H Shi, AR Pattnaik, **CA Aguila**, ..., EC Conrad (2024). Utility of intracranial EEG networks depends on re-referencing and connectivity choice. *Brain Communications*.
- WKS Ojemann, BH Schied, **CA Aguila**, ..., B Litt (2023). Resting-state background features demonstrate multidien cycles in long-term EEG device recordings. *Brain Stimulation*.
- V Mayo, C Alver, **CA Aguila**, ..., A Agarwal (2020). Modular Tissue Engineering of Bone Microenvironment as an Extravasation Model. (*in preparation*).
- DL Jones, AW Carrico, **CA Aguila**, ..., M Kumar (2018). Methamphetamine-associated dysregulation of the hypothalamic-pituitary-thyroid axis. *Journal of behavioral medicine*.

CONFERENCE PRESENTATIONS

- K Nix, K Walsh, **CA Aguila**, ..., B Litt. Network Characteristics and Treatment Response in Epilepsy: Responsive Neurostimulation Non-Responders exhibit more focal Interictal Spiking. Society for Neuroscience 2025. San Diego, CA, Dec. 2025.
- CA Aguila**, S Zhu, S Lavelle, ..., EC Conrad. Low Frequency stimulation has heterogenous effects on interictal spike rates. International Conference for Technology and Analysis of Seizures (ICTALS) 2025. Montreal Canada, June 2025
- S Lavelle, WKS Ojemann, **CA Aguila**, ..., N Sinha. Functional and structural networks drive cortico-cortical evoked potentials. International Conference for Technology and Analysis of Seizures (ICTALS) 2025. Montreal Canada, June 2025
- WKS Ojemann, **CA Aguila**, ..., B Litt. Predicting network reorganization induced by high frequency stimulation. International Conference for Technology and Analysis of Seizures (ICTALS) 2025. Montreal Canada, June 2025
- WKS Ojemann, Z XU, **CA Aguila**, ..., EC Conrad. Neural dynamic divergence: an unsupervised machine learning approach to modelling ictal state transitions. AI in Epilepsy 2025. Breckenridge, CO, March 2025
- CA Aguila**, A Lucas, AR Pattnaik, ..., EC Conrad. Spatial Patterns of Spread and Morphology of Seizures and Interictal Spikes Identify Patients with Mesial Temporal Lobe Epilepsy. American Epilepsy Society 2024. Los Angeles, CA, December 2024
- S Lavelle, AR Pattnaik, **CA Aguila**, ..., B Litt. Combining Spatial Sampling of Intracranial EEG and Ictal Epileptogenicity to Predict Focality. American Epilepsy Society 2024. Los Angeles, CA, December 2024
- WKS Ojemann, AR Pattnaik, **CA Aguila**, ..., EC Conrad. Stimulation-induced Seizures Map Spontaneous Seizure Spread Networks. American Epilepsy Society 2024. Los Angeles, CA, December 2024
- I Ong, AR Pattnaik, **CA Aguila**, ..., B Litt. Normative Intracranial EEG Localizes Epileptic Abnormalities Across States of Wakefulness. American Neurological Association Annual Meeting. Orlando, FL, September, 2024.
- CA Aguila**, A Lucas, AR Pattnaik, ..., EC Conrad. Role of Morphology in Interictal Spikes to Elucidate Seizure Generation. American Epilepsy Society 2023. Orlando, FL, December 2023
- WKS Ojemann, BH Schied, **CA Aguila**, ..., B Litt. Multidien Cycles in Interictal Background Features of Long-Term EEG Recordings. American Epilepsy Society 2023. Orlando, FL, December 2023
- AR Pattnaik, IZ Ong, **CA Aguila**, ..., B Litt. Quantitative Intracranial EEG Abnormalities Across States of Consciousness in Drug-resistant Epilepsy. American Epilepsy Society 2023. Orlando, FL, December 2023
- CA Aguila**, A Lucas, AR Pattnaik, ..., EC Conrad. Morphological Interictal Spike Analysis for Seizure Onset Zone Localization. American Epilepsy Society 2022. Nashville, TN, December 2022
- CA Aguila**, A Lucas, AR Pattnaik, ..., EC Conrad. Morphological Interictal Spike Analysis for Seizure Onset Zone Localization. Biomedical Engineering Society 2022 Annual Meeting. San Antonio, TX, October 2022
- CA Aguila**, M Ishahak, A Agarwal. Characterization of Shear Stress in an Organ-on-Chip Platform. Biomedical Engineering Society 2019 Annual Meeting. Philadelphia, PA, October 2019

TEACHING EXPERIENCE

Graduate Teaching Fellow

January 2023 - Present

BE 5210 – Brain Computer Interfaces

Department of Bioengineering, University of Pennsylvania